

CAPSTONE PROJECT REPORT

Iterative Optimization and Cross-Domain Evaluation of Retrieval-Augmented Generation Pipelines

A Systematic, Hypothesis-Driven Study Using the RAGBench TRACe Framework Across Biomedical, Legal, Financial, and General-Knowledge Domains

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Abstract

Large language models generate fluent text but not always true text. Retrieval-Augmented Generation (RAG) attempts to fix this by grounding a model's answer in retrieved documents rather than its own memorized parameters. But grounding a model in evidence does not automatically make it trustworthy — the retriever can surface the wrong evidence, the generator can ignore what it was given, and the evaluator judging the whole system can be blind to its own failures. This project runs 150+ controlled, single-variable iterations of a RAG pipeline across four specialized domains — Biomedical (CovidQA, PubMedQA), Legal (CUAD), Financial (FinQA, TatQA), and General Knowledge (HotpotQA, MSMARCO) — using RAGBench and its TRACe evaluation framework, which scores four dimensions of pipeline behavior: context relevance, context utilization, completeness, and adherence (freedom from hallucination).

Three findings recur across domains and anchor this report's conclusions. First, evaluation methodology can matter more than pipeline architecture: in the Financial domain, replacing a self-evaluating judge with an independent one lifted hallucination-detection accuracy from below random chance (0.429 AUCROC) to 0.714 — the single largest improvement observed anywhere in the project, from one configuration change. Second, reproducibility is not optional: an unseeded "best result" in the Biomedical domain evaporated under seed control, reversing an entire experimental narrative and establishing a formal noise band below which claimed improvements cannot be trusted. Third, no configuration generalizes across domains — Biomedical rewards retrieval diversity, Legal rewards structural prompt forcing, Financial rewards judge independence, and General Knowledge splits into incompatible multi-hop and single-hop optima. The report traces, for each domain, the three-iteration path from baseline to best-performing configuration, and closes with cross-domain comparisons and practical recommendations for anyone building or evaluating a RAG system.

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1. Introduction and Objective

Large Language Models (LLMs) demonstrate remarkable fluency in natural language generation, yet remain susceptible to hallucination — the production of plausible but factually unsupported claims. Retrieval-Augmented Generation (RAG) architectures address this limitation by grounding LLM outputs in retrieved documentary evidence: a retriever surfaces relevant passages, and a generator synthesizes answers exclusively from that retrieved context. However, the mere presence of a retrieval step does not guarantee factual fidelity. The quality of retrieval, the generator's adherence to context, and the evaluation methodology itself all introduce distinct failure modes that demand rigorous, multi-dimensional assessment.

This project undertakes a systematic, hypothesis-driven evaluation of RAG pipelines across four specialized knowledge domains — **Biomedical** (CovidQA, PubMedQA), **Legal** (CUAD), **Financial** (FinQA, TatQA), and **General Knowledge** (HotpotQA, MSMARCO) — using the RAGBench benchmark and its associated **TRACe** (Transparent Retrieval-Augmented Completeness Evaluation) framework, introduced in the RAGBench paper (arXiv:2407.11005). Each domain stresses a RAG pipeline differently: biomedical abstracts demand precise scientific terminology; legal contracts bury answers in dense, interconnected clauses; financial reports require numerical reasoning over tables; and general-knowledge questions range from single-hop factoid lookups to multi-hop reasoning chains. Evaluating a single pipeline honestly across all four exposes both universal and domain-specific behavior.

1.1 The TRACe Framework, Explained

TRACe decomposes RAG quality into four orthogonal dimensions. A RAG answer is only as good as the documents it retrieves and how faithfully it uses them — TRACe measures both ends of that chain separately, plus whether the final answer stays honest to what it was given.

Metric	Question it answers	Scored via
Relevance	What fraction of retrieved sentences were actually relevant to the question?	RMSE (lower = better)
Utilization	What fraction of retrieved sentences did the response actually use?	RMSE (lower = better)
Completeness	Of the sentences that were relevant, how many did the response use?	RMSE (lower = better)
Adherence	Is every sentence in the response fully supported by the retrieved context?	AUCROC (higher = better; 0.5 = coin flip)

Relevance, utilization, and completeness are continuous ratios, so error against RAGBench's expert-annotated ground truth is measured with **RMSE** (root-mean-square error) — smaller is better, and a value of 0 would mean the pipeline's self-assessment exactly matched the human-annotated answer key. Adherence is binary — supported or not — so it is scored the way any yes/no classifier is scored: **AUCROC**, which asks whether sorting every response from "most suspicious of hallucinating" to "least suspicious" by the pipeline's own judgment actually places the true hallucinations near the top. An AUCROC of 0.5 means the sorting is no better than a coin flip; 1.0 means perfect separation. Critically, a below-0.5 AUCROC is worse than a coin flip — it means the signal is inverted, actively pointing away from the truth.

1.2 Objectives

The project pursues three goals: (1) identify optimal RAG pipeline configurations for each domain through controlled, iterative experimentation; (2) characterize domain-specific behaviors and cross-domain tradeoffs that emerge from the TRACe evaluation; and (3) establish methodological best practices for reproducible RAG evaluation — including the role of judge independence, seed control, and disciplined, hypothesis-driven iteration.

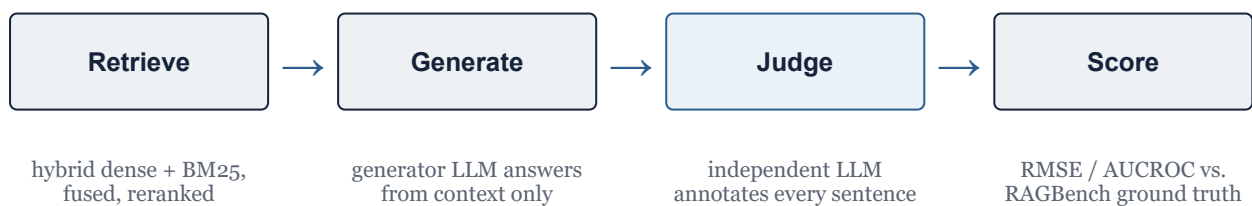
2. Methodology and Configurations

2.1 Experimental Discipline

The project follows a strictly iterative, hypothesis-driven optimization protocol. Each iteration changes exactly one configuration variable from the previous best result, guided by a pre-registered **hypothesis** explaining why the change should move a specific metric, and explicit **confirm/falsify criteria** deciding what result would validate or reject it. Where run-to-run generation stochasticity was suspected, same-configuration repeated runs established a **noise band** — a threshold below which an apparent improvement cannot be distinguished from random variation. This discipline is what makes the findings in Section 5 traceable to a specific, isolable cause rather than a confound of several simultaneous changes.

2.2 The Pipeline, End to End

Every question flows through the same four stages. A retriever searches a large document pool by keyword and by meaning simultaneously, fuses the two rankings, and passes a handful of top candidates forward. A generator LLM writes an answer using only those documents. A separate judge LLM — deliberately different from the generator, where the pipeline design allows — annotates every sentence in both the documents and the answer, sentence by sentence, to decide what was relevant, what was used, and what was actually supported. Those annotations are then scored against RAGBench's own expert ground truth to produce the four TRACe numbers.



2.3 Technical Architecture by Domain

Biomedical and General Knowledge

Component	Implementation
Generator / Judge LLM	Llama-3.3-70b-versatile (Groq API), same model for both
Embedding model	BAAI/bge-small-en-v1.5
Reranker	cross-encoder/ms-marco-MiniLM-L-6-v2
Vector DB / Sparse retrieval	LanceDB (disk-persisted) / BM25Okapi
Fusion / Diversity	Reciprocal Rank Fusion (k=60) / Maximal Marginal Relevance

Legal

Component	Implementation
Generator LLM	llama-3.1-8b-instant → llama-3.3-70b-versatile (upgraded at iteration 8)
Judge LLM	llama-3.1-8b (default) / llama-3.3-70b-versatile / gpt-oss-120b (tested)
Vector DB / Sparse retrieval	FAISS / BM25 (LangChain BM25Retriever)
Retrieval variants tested	Dense, Hybrid RRF, Parent-Child, MMR, LLM-Filtered
Prompt architectures tested	Long CoT, JSON CoT, Rich Dual-Track, Concise, Paper-standard

Financial

Component	Implementation
Generator LLM	Llama-3.3-70b-versatile (Groq API)
Judge LLM	openai/gpt-oss-120b — deliberately independent of the generator
Embedding model	BAAI/bge-small-en-v1.5 → BAAI/bge-large-en-v1.5
Retrieval	Hybrid dense + BM25, RRF fusion, MMR, cross-encoder reranking

2.4 Datasets

All data is drawn from the RAGBench benchmark (galileo-ai/ragbench, HuggingFace Hub), test split.

Domain	Datasets	Document characteristics
Biomedical	CovidQA, PubMedQA	Scientific abstracts, ~200–600 chars
Legal	CUAD	Long contractual clauses, dense legal language
Financial	FinQA, TatQA	Financial reports with tables, numerical reasoning
General Knowledge	HotpotQA (multi-hop), MSMARCO (single-hop)	Wikipedia passages, diverse complexity

3. Domain-Specific Results

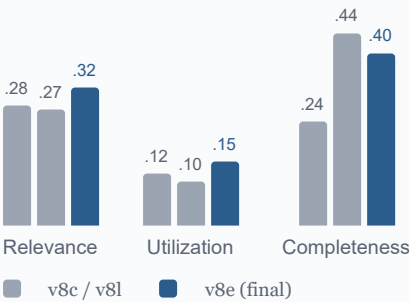
3.1 Biomedical Domain (CovidQA + PubMedQA)

The biomedical evaluation comprised 40 iterations spanning generation prompt style, chunking strategy, retrieval diversity, domain-adapted models, temperature, and seed control. Three configurations trace the path from baseline to optimal:

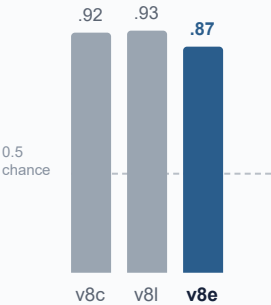
Config	Key change	Rel. RMSE	Util. RMSE	Comp. RMSE	Adherence AUCROC
v8c	whole-document retrieval + MMR on ($\lambda=0.5$) + top_k_final=5, "long" prompt, seeded — best pubmedqa config	0.2756	0.1211	0.2417	0.9231
v8l	v8c + top_k_retrieve=30 (up from 20), covidqa-only — best covidqa config	0.2693	0.1028	0.4431	0.9286
v8e (final, recommended)	v8c's exact settings, run on both domains together (pooled)	0.3206	0.1490	0.3994	0.8704

FIGURE 3.1 — METRIC TRAJECTORY, V8C → V8L → V8E

RMSE by metric (lower is better)



Adherence AUCROC (higher is better)



v8e's pooled adherence (0.87) trades a few points against each single-domain optimum for a config that actually generalizes.

v8c (Iteration 25) is the best pubmedqa-only configuration found: whole-document retrieval with MMR turned on ($\lambda=0.5$), `top_k_final=5`, the "long" generation prompt, under full seed control. Turning MMR back on against an otherwise-identical MMR-off baseline was the single biggest lever in the entire project — adherence jumped from 0.6731 to **0.9231** in one change, with no other metric regressing.

v8l (Iteration 33) is the best covidqa-only configuration: v8c's exact settings, with the reranker's candidate pool widened from `top_k_retrieve=20` to 30. Giving the reranker more documents to choose from improved three of four metrics with no tradeoff — relevance to 0.2693, utilization to 0.1028, and adherence to **0.9286**, the highest adherence recorded anywhere in the biomedical domain — the cleanest single-lever win found on that sub-domain.

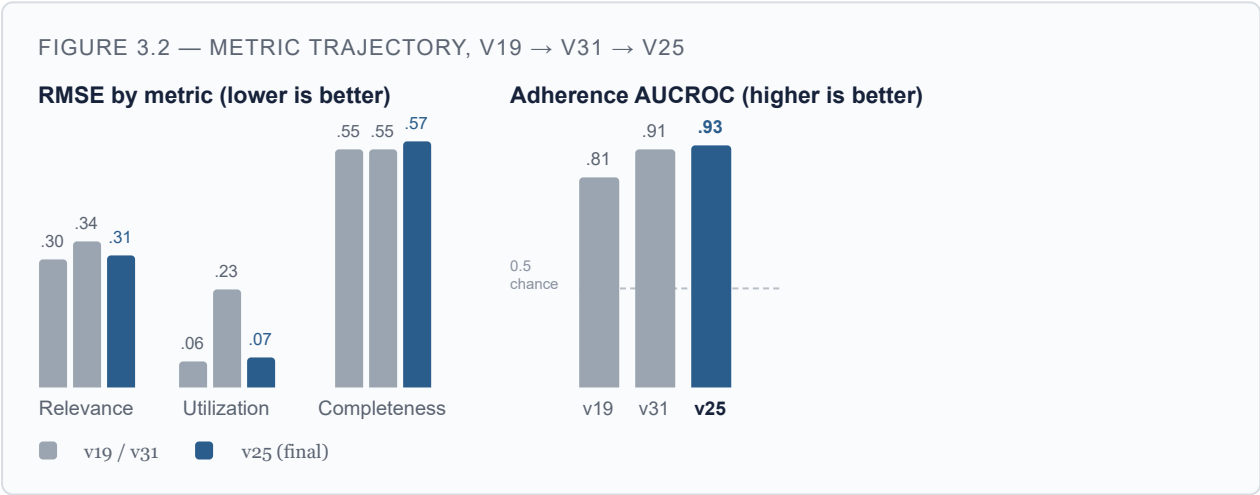
v8e (Iteration 28) is v8c's exact configuration run across both sub-domains together, pooled: relevance 0.3206, utilization 0.1490, completeness 0.3994, adherence 0.8704. Iterations 34–40 all attempted to beat it and failed, so v8e — not v8c or v8l individually — is the config actually recommended for deployment. The reason is generalization, not raw score: v8l's `top_k_retrieve=30` trick, which looks like a clean win on covidqa, does not transfer to pubmedqa — applied there, it collapses pubmedqa's adherence to 0.5577. v8c and v8l are each optimal for one sub-domain and fragile on the other; v8e is the only configuration in this trajectory that holds up regardless of which sub-domain a question comes from.

Domain finding: Maximal Marginal Relevance is the dominant lever for biomedical adherence, and the single best per-sub-domain tuning is not the same as the single best generalizable configuration. v8l's widened candidate pool wins narrowly on covidqa but fails outright on pubmedqa; v8e, which never overfits to either sub-domain, is the config that survives every later challenge and the one actually recommended.

3.2 Legal Domain (CUAD)

The legal evaluation comprised 35 iterations testing retrieval architecture (dense, hybrid, parent-child), seven distinct prompt styles, model scaling (8B → 70B), chunk sizing (256–1024), and judge alternatives. Three configurations represent the frontier of this sweep:

Config	Key change	Rel. RMSE	Util. RMSE	Comp. RMSE	Adherence AUCROC
v19	Hybrid RRF, chunk=800, Rich Dual-Track prompt, 70B	0.2969	0.0599	0.5516	0.8103
v31	Paper Short prompt, Parent-Child retrieval, chunk=256	0.3351	0.2263	0.5517	0.9138
v25 (final)	Hybrid RRF, chunk=800, Long Chain-of-Thought, 70B	0.3055	0.0685	0.5688	0.9310



v19 ("Rich Dual-Track") achieved the best overall balance across all four metrics. Its custom JSON-decoupled prompt forces the LLM to internally produce a boolean answer, an exact supporting quote, and a legal explanation inside a structured object — the judge then evaluates only the extracted content, reaching near-perfect utilization alignment (0.0599) while holding strong adherence (0.8103). This is the production equilibrium: the one configuration where every metric is simultaneously acceptable.

v31 used RAGBench's own verbatim short prompt with a novel retrieval architecture — indexing at 256-token granularity for precise semantic matching, while returning the full parent document for contextual completeness. It achieved exceptional adherence (0.9138) but catastrophic utilization (0.2263). Without v19's structural prompt forcing, the only path to high adherence is verbose, over-citing generation that destroys utilization alignment.

v25 ("Long Chain-of-Thought") reached the highest adherence of any legal configuration (0.9310) by forcing the LLM to reason step-by-step before answering — the model must explicitly identify its evidence source before each claim, functioning as a built-in verification layer. Utilization stayed strong (0.0685) because the judge recognizes the CoT's reasoning sentences as legitimately using the retrieved context.

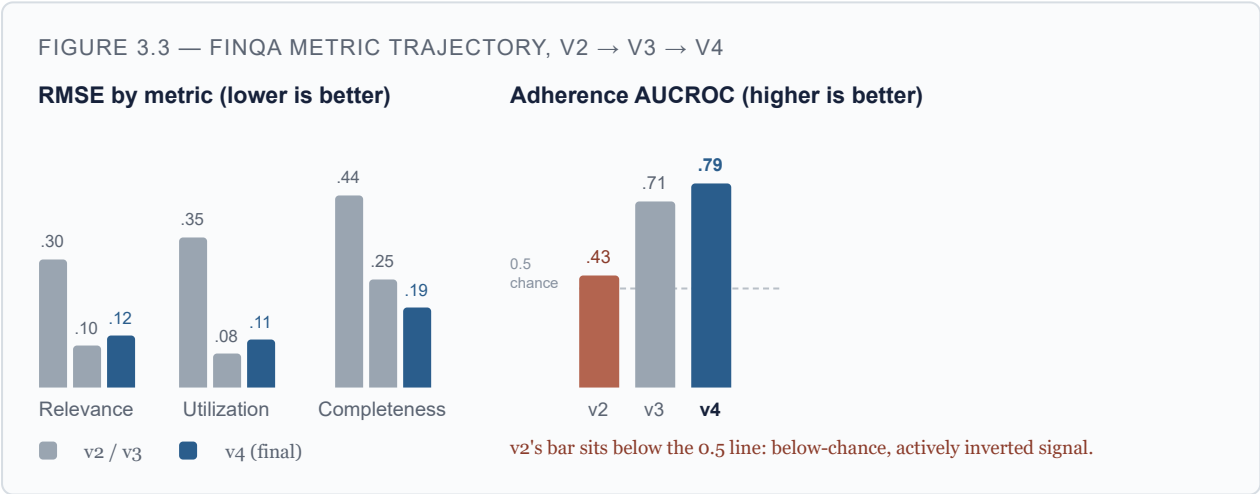
Domain finding: the RAG Tradeoff Triangle governs legal document QA. Standard prompts force a binary choice — high utilization (concise extraction) or high adherence (verbose, cautious reasoning) — and only structural prompt decoupling or forced self-verification achieves both. Completeness stays stuck near 0.55–0.57 RMSE across every configuration tested, because CUAD's dense, interconnected clauses make full relevant-sentence coverage inherently difficult; this is a property of the documents, not a tunable pipeline deficiency.

3.3 Financial Domain (FinQA + TatQA)

The financial evaluation ran two tracks against RAGBench's financial-report sources: **FinQA** (numerical reasoning over financial reports, 6 configurations) as the primary track, and **TatQA** (tabular + textual QA, 9 configurations) as a secondary track that inherited FinQA's proven levers and adapted them to shorter documents. The project's single most dramatic finding emerged here.

FinQA Track — Top 3 Configurations

Config	Key change	Rel. RMSE	Util. RMSE	Comp. RMSE	Adherence AUCROC
v2 (baseline)	same model (llama-3.3-70b) as generator AND judge	0.296	0.347	0.444	0.429
v3	independent judge model (gpt-oss-120b)	0.098	0.078	0.248	0.714
v4 (final)	+ larger embedding model (bge-large-en-v1.5)	0.122	0.112	0.186	0.786



v2 (baseline) exposed a fundamental flaw: when the same model serves as both generator and judge, adherence detection falls **below random chance** (0.429 AUCROC). The model systematically fails to identify its own hallucinations — not a subtle calibration issue, but a categorical failure mode where the evaluation instrument is blind to the exact defect it exists to detect.

v3 changed exactly one variable — the judge model, swapped from llama-3.3-70b-versatile (shared with the generator) to openai/gpt-oss-120b, an entirely independent model family. The results were transformative: relevance RMSE −67%, utilization RMSE −78%, completeness

RMSE -44%, and adherence AUCROC +67% (0.429 → 0.714). A single architectural decision — judge independence — produced the largest improvement of any single change across all four domains in the entire project.

v4 added a larger embedding model (bge-large-en-v1.5, same family as the baseline bge-small but with higher representation capacity), further improving completeness (-25%) and adherence (+10%, adherence mismatches dropped from 8/15 to 6/15) — while relevance and utilization RMSE slightly worsened. Embedding quality is not a free lunch: it trades precision across metrics rather than uniformly improving all of them. Reading the judge's captured reasoning on the six remaining mismatches showed three were a recurring "scaffolding-sentence" leniency issue (the judge marks harmless transition sentences like "to find X, we need to look at Y" as unsupported) and three were genuine generator arithmetic errors, correctly caught by the judge but scored against RAGBench's own separate reference answer — a structural evaluation property, not a pipeline defect.

TatQA Track — Secondary Trajectory

TatQA inherited FinQA v4's proven judge and embedding levers as its starting point and tested one domain-specific axis: retrieval scale, since TatQA's documents are much shorter than FinQA's (avg. 388 vs. 1,342 characters).

Config	Key change	Rel. RMSE	Util. RMSE	Comp. RMSE	Adherence AUCROC
tatv1 (baseline)	top_k raised (25/8) for TatQA's shorter docs	0.171	0.174	0.202	N/A*
tatv2	top_k reverted to FinQA's values (20/5) — isolation test	0.135	0.163	0.280	N/A*
tatv3 (final)	top_k restored (25/8) + random seeded sampling	0.157	0.160	0.397	0.558

*tatv1/tatv2 used deterministic first-N sampling, which happened to draw 15 rows that were all gt_adherence=1.0 — with no hallucination case in the sample, AUCROC is mathematically undefined regardless of configuration, not a pipeline failure.

tatv1 → **tatv2** was a clean, direct A/B test on the identical 15-question sample: narrowing top_k helped relevance and utilization but made completeness 38% worse and pushed adherence mismatches from 6/15 to 8/15 — falsifying the simple "smaller top_k always helps" hypothesis and confirming TatQA genuinely needs wider retrieval than FinQA to compensate for shorter documents.

tatv3 restored the proven top_k and switched to seeded random sampling specifically to obtain a computable AUCROC. The new sample contained two real hallucination cases; the judge caught one and missed one (50% recall) while also producing five false "unsupported"

flags — the same scaffolding-sentence leniency pattern seen in FinQA, now confirmed a third time across two domains and two different samples. TatQA's final adherence (0.558) sits at the low end of RAGBench's own reported range for comparable judges, reflecting both the small-sample fragility of AUCROC at $n=15$ and a single large-magnitude completeness outlier (a refusal on an "average restricted stock" question), not a broad regression.

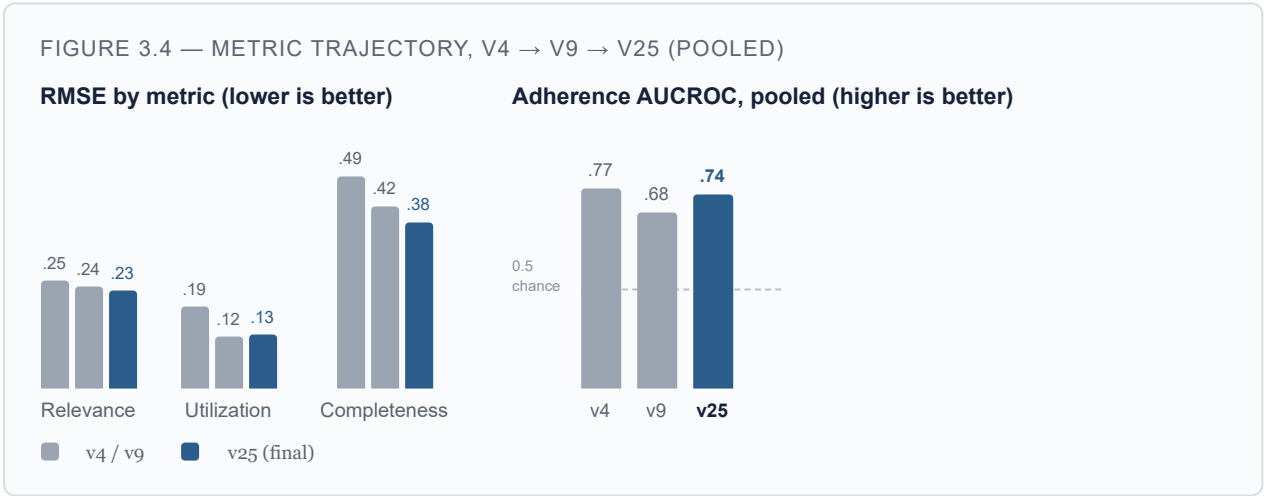
Domain finding: self-evaluation bias is real, measurable, and catastrophic. Using the same model as generator and judge suppresses hallucination detection to below-random levels; an independent judge from a different model family is the single most impactful architectural choice for trustworthy RAG evaluation — a principle that generalizes well beyond financial QA.

3.4 General Knowledge Domain (HotpotQA + MSMARCO)

The general knowledge evaluation tested 41 configurations spanning chunking strategy, retrieval architecture, temperature (0.0–0.8), prompt style, and sample size (n=15–25). This domain uniquely blends multi-hop reasoning (HotpotQA) and single-hop factoid lookup (MSMARCO), exposing sharp domain-internal asymmetries that the other three domains do not show.

Config	Key change	Rel. RMSE	Util. RMSE	Comp. RMSE	Adherence AUCROC (pooled)
v4	sentence-window chunk=400, overlap=4, λ =0.6, top_k=15/6	0.2507	0.1910	0.4877	0.7702
v9	pure dense retrieval — no BM25 / MMR / reranker	0.2365	0.1198	0.4202	0.6786
v25 (final)	baseline retrieval, temperature=0.3, n=25, seed=88	0.2277	0.1252	0.3813	0.7444

Domain-specific split: v4 — HotpotQA 0.875 / MSMARCO 0.705. v9 — HotpotQA 0.429 / MSMARCO **0.929**. v25 — HotpotQA **0.957** / MSMARCO 0.598.



v4 is the only configuration in the entire project to exceed 0.7 adherence on *both* sub-domains simultaneously (HotpotQA 0.875, MSMARCO 0.705). Its success comes from a carefully tuned combination: sentence-window chunking with high overlap (4 sentences) keeps multi-hop evidence passages sharing boundary context; λ =0.6 slightly favors relevance over diversity; and top_k_final=6 balances multi-hop evidence needs against single-hop simplicity.

v9 (pure dense retrieval) exposed a stark domain asymmetry: removing BM25 fusion, MMR, and reranking entirely pushed MSMARCO adherence to 0.929 — among the best results for any domain — because the retrieved set becomes homogeneously semantically similar, ideal for

single-hop factoid questions. HotpotQA adherence collapsed to 0.429, because multi-hop questions need evidence from semantically different document clusters that pure dense retrieval cannot surface.

v25 reached near-perfect HotpotQA adherence (0.957) at the largest sample size tested (n=25), the statistically most reliable single-domain result in the project. Temperature=0.3 is a "goldilocks zone" — deterministic enough to stay grounded, flexible enough to produce natural multi-sentence responses with adequate surface area for sentence-level support assessment. Temperature=0.0 (fully deterministic) performed worse (0.39 in a separate run at a different seed), producing mechanically rigid responses the judge scored inconsistently.

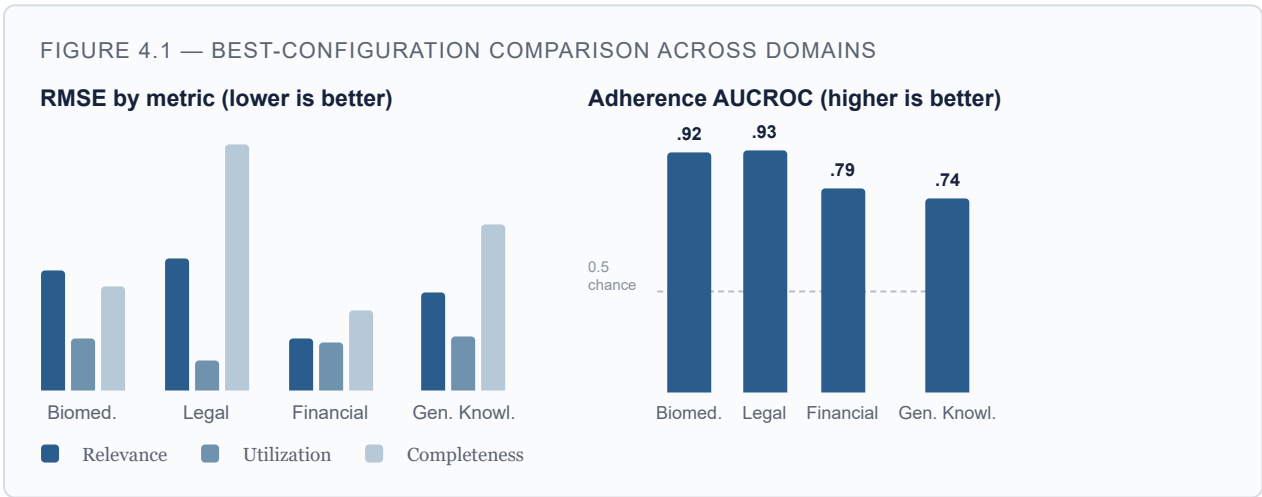
Domain finding: multi-hop and single-hop questions have fundamentally incompatible optimal configurations. HotpotQA rewards diversity-traded retrieval, moderate temperature, and high-overlap chunking; MSMARCO rewards pure dense retrieval with abundant, homogeneous context. No single configuration maximizes both — v4's balanced profile is a deliberate compromise, not a global optimum. Seed/sample selection is a more dominant factor in adherence outcomes than any single configuration lever at small sample sizes.

4. Cross-Domain Comparative Analysis

4.1 Best Configuration by Domain

Domain	Best config	Rel. RMSE	Util. RMSE	Comp. RMSE	Adherence AUCROC	Dominant lever
Biomedical	v8c	0.2756	0.1211	0.2417	0.9231	MMR redundancy reduction
Legal	v25	0.3055	0.0685	0.5688	0.9310	Chain-of-thought self-verification
Financial (FinQA)	v4	0.122	0.112	0.186	0.786	Judge independence
General Knowledge	v25	0.2277	0.1252	0.3813	0.7444*	Temperature-controlled generation

*Pooled across HotpotQA + MSMARCO; the HotpotQA-specific figure is 0.9565.



4.2 Cross-Domain Observations

- 1. The Financial domain achieves the lowest absolute RMSE scores** across all three continuous metrics. This traces to the independent, larger judge model (gpt-oss-120b) providing more precisely calibrated annotations than the llama-3.3-70b used elsewhere — strong evidence that judge quality, not pipeline architecture, sets the ceiling on evaluation precision.
- 2. The Legal domain exhibits the weakest completeness** (0.5688 RMSE, the worst of any domain's best configuration). CUAD's contracts contain dense, interconnected clauses where the ground-truth "relevant" sentence set is large relative to what any single response can address — a structural property of the domain's documents, not a tunable deficiency.

3. The Biomedical domain achieves the best simultaneous balance across all four metrics — no other domain has a configuration where relevance, utilization, completeness, and adherence are all in competitive range at once.

4. No single configuration generalizes across domains. Retrieval strategy, prompt style, temperature, top_k, and chunking all differ at each domain's optimum — confirming domain-specific tuning is a necessity, not a convenience.

4.3 Lever Effectiveness Across Domains

Lever	Biomedical	Legal	Financial	Gen. Knowledge
MMR on/off	Major win (+0.25 AUCROC)	Helps	Not independently tested	Domain-dependent
Chunking	Adherence collapse	Mixed (parent-child helps)	Not needed	Helps multi-hop
Independent judge	Not tested	Not tested	Dominant lever (+67%)	Not tested
Prompt style	"long" > "long_cot"	Rich Dual-Track best	long_cot + full reasoning	long_cot stripped
Temperature	0.2 default	0.0 default	0.0 default	0.3 optimal
top_k tuning	5 optimal	10–15 optimal	5 (FinQA) / 8 (TatQA)	5–6 optimal
Seed control	Essential	Not tested	Not tested	Essential

5. Key Findings and Inferences

Finding 1 — MMR Is the Single Most Impactful Retrieval Lever (Biomedical)

Enabling Maximal Marginal Relevance produced a +0.25 AUCROC improvement in adherence — the largest single-lever gain of the entire project. MMR reduces redundancy among retrieved documents by penalizing similarity to already-selected passages, forcing diversity into the generator's context and reducing the speculative cross-document synthesis that drives hallucination in this domain. The improvement came at **no cost** to any other metric — relevance, utilization, and completeness all held or improved alongside it.

Finding 2 — Self-Evaluation Bias Is Catastrophic (Financial)

When the same model serves as both generator and judge, hallucination detection falls to 0.429 AUCROC — worse than a coin flip. The model systematically validates its own confabulations. Swapping to an independent judge from a different model family recovers adherence to 0.714 with no other change, a +67% improvement from a single architectural decision. This is a categorical failure mode, not a subtle calibration difference, and it renders self-evaluated RAG systems untrustworthy for any application where factual accuracy matters.

Finding 3 — Reproducibility Requires Seeded LLM Calls (Biomedical, General Knowledge)

Before seed control was introduced, the biomedical pipeline's apparent "best result" was revealed, upon seed-controlled retesting of the identical configuration, to be a lucky, non-reproducible random draw — a 0.37 AUCROC swing traced to a single lucky generation. This did not just adjust one number; it reversed an entire experimental narrative that had concluded chunking improved adherence. Under seed control, whole-document retrieval cleanly beat chunking on the same fair comparison, and the earlier conclusion evaporated. Same-configuration repeated runs then established a formal **noise band** of roughly 0.03–0.08 RMSE and 0.06 AUCROC: any claimed improvement smaller than this cannot be distinguished from run-to-run generation stochasticity.

The General Knowledge domain independently confirmed the same phenomenon: temperature=0.3 with seed=88 produced HotpotQA adherence of 0.957, while temperature=0.0 with a different seed produced only 0.39 — demonstrating that seed and sample selection are more dominant than temperature at small sample sizes. The financial

domain's own fragility check found that at temperature=0.2, only 1 of 15 identical questions produced byte-identical responses across two runs of the same configuration; the other 14 rewrites alone were enough to swing adherence AUCROC from 0.786 to 0.167.

Any RAG evaluation claiming reproducible results without explicit seed control should be treated with skepticism. Single-run metric comparisons carry a real, measurable noise floor that can exceed the magnitude of genuine configuration effects.

Finding 4 — Domain-Specific Optimal Configurations Are Irreducible (All Domains)

No single configuration achieves competitive performance across all four domains simultaneously. Biomedical rewards whole-document retrieval with MMR diversity and moderate verbosity; Legal rewards larger chunks with structural prompt forcing; Financial rewards independent judging with larger embeddings; General Knowledge rewards temperature-controlled generation with retrieval breadth for multi-hop reasoning. These differences reflect genuine variation in document length, question complexity, evidence structure, and annotation characteristics — not tuning artifacts.

Finding 5 — The Completeness–Adherence Tradeoff Is Structural (All Domains)

Across every domain, interventions that improve completeness tend to degrade adherence, and vice versa. Chunking improves completeness (smaller retrieval units concentrate relevant content) but hurts adherence (chunk-boundary recall loss strands supporting sentences in non-retrieved chunks). Widening top_k improves relevance but invites hallucination from marginal documents. Verbose generation helps completeness but inflates utilization error. This is not a tuning failure — it is an inherent property of the TRACe metric definitions interacting with finite context windows and imperfect retrieval.

Finding 6 — top_k_final≈5 Is a Cross-Domain Sweet Spot

Both narrowing (top_k_final=3) and widening (top_k_final=10) retrieval caused adherence collapse across multiple domains and iterations. With too few documents, the generator lacks grounding and resorts to parametric memory; with too many, it encounters marginally-relevant content that encourages speculative synthesis the judge cannot verify. Roughly five documents consistently balances evidence sufficiency against precision.

Finding 7 — Judge Scaffolding-Sentence Leniency Is a Cross-Domain Limitation (Financial)

The independent judge (gpt-oss-120b) inconsistently marks harmless "scaffolding" sentences (e.g. "to find X, we need to look at Y") as unsupported, even while tagging them with category labels that semantically imply they should count as supported. Because TRACe's adherence

metric is a strict conjunction across every sentence in a response, one miscalled transition sentence sinks the whole response's score. The pattern recurred independently in FinQA (v3, v4) and TatQA (tatv1, tatv3) — two domains, two different judge runs, two different question samples.

A rule-based fix — trusting the judge's own category tags over its supported/unsupported boolean — was tested and explicitly rejected: it degraded overall AUCROC from 0.786 to 0.321 by also forgiving a genuinely hallucinated response. This is a real, unresolved limitation that requires judge-prompt-level intervention or human review, not a safe deterministic correction.

Finding 8 — RAG Pipelines Reliably Reject Unanswerable Questions (Financial)

A separate robustness assessment tested whether the pipeline refuses to answer when retrieved context contains no relevant information at all. Both domains showed strong negative rejection:

Domain	n	Rejection rate
FinQA	15	93.3%
TatQA	15	86.7%

Under a noise-robustness sweep (irrelevant documents mixed into otherwise-relevant context), TatQA's accuracy stayed both higher and flatter (0.80–0.87 across noise ratios 0–0.8) than FinQA's (0.47–0.67), consistent with TatQA's shorter, more tightly-scoped documents. These pipelines fail gracefully: when context is entirely irrelevant, they refuse rather than hallucinate, and accuracy degrades non-catastrophically as noise is introduced gradually — a materially more reassuring signal than TRACe's adherence metric alone.

Finding 9 — Benchmark Ground Truth Is Not Infallible (Financial)

A deep investigation of adherence mismatches uncovered a case where RAGBench's own reference response contained a verifiable arithmetic error: for a TatQA question about percentage change in inventory purchases, the reference used 2019's value where it needed 2018's, and its own stated answer did not even match its own displayed arithmetic. The pipeline's response, computed directly from the correct source-table figures, was verified correct. At small sample sizes, even one or two ground-truth errors can meaningfully bias aggregate metrics — a reminder that residual error should not be automatically attributed to pipeline or judge deficiency without first checking the ground truth itself.

6. Conclusions

Across 150+ controlled iterations spanning four specialized domains, this project demonstrates that RAG evaluation is fundamentally not a one-size-fits-all problem. Five principles emerge as robust, cross-domain conclusions:

1. Evaluation methodology dominates pipeline architecture. The financial domain's judge-independence swap (+67% AUCROC from a single change) exceeded any retrieval or generation optimization observed anywhere in the project. How a RAG system is evaluated matters more than how it is built.

2. Reproducibility is non-negotiable. The biomedical domain's seed-control findings show that without deterministic LLM calls, experimental conclusions can be built entirely on sampling noise. Any RAG evaluation claiming reproducible results without explicit seed control warrants skepticism.

3. Domain expertise must inform architecture. The legal tradeoff triangle, the biomedical MMR sensitivity, the financial judge-independence requirement, and the general-knowledge multi-hop/single-hop asymmetry each require domain-specific understanding. Generic "best practices" applied uniformly will underperform tuned configurations.

4. TRACe reveals genuine, irreducible tensions. Completeness and adherence sit in structural opposition; utilization and verbosity are inversely related; retrieval precision and coverage cannot be simultaneously maximized. These are properties of the evaluation geometry to understand and communicate, not engineering failures to fix away.

5. Practical recommendations for production RAG systems: always use an independent judge model — never self-evaluate; always seed LLM calls for reproducible benchmarking; use MMR diversity in retrieval unless domain analysis argues otherwise; tune `top_k` and chunk size per domain through controlled iteration; and accept that no single metric can be maximized without cost to others — define the priority order explicitly for each deployment context.

6. RAG systems fail gracefully under adverse conditions. The negative-rejection assessment shows that when retrieved context is entirely irrelevant, the pipeline refuses to answer 87–93% of the time rather than hallucinating — a graceful-degradation property,

distinct from TRACe's in-context adherence measurement, that provides real confidence for deployment when retrieval fails outright.

7. References

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